Department of Mathematics, IIT Madras
MA2031-Linear Algebra for Engineers
Assignment-V

1. Fix a vector u in an ips V. Define a linear functional T on V by $Tv = \langle v, u\rangle$ for $v \in V$. What is $T^*(\alpha)$ for a scalar α?
2. Define an operator T on $\mathbb{F}^n$ by $T(a_1, a_2, \ldots, a_n) = (0, a_1, \ldots, a_{n-1})$. What is $T^*(a_1, \ldots, a_n)$?
3. Prove that a linear operator T on a finite dimensional ips is invertible iff T^* is invertible. In that case, show that $(T^*)^{-1} = (T^{-1})^*$.
4. Prove the following for a finite dimensional inner product space:
$V = N(T) \oplus N(T)^\perp = R(T) \oplus R(T)^\perp = N(T^*) \oplus N(T^*)^\perp = R(T^*) \oplus R(T^*)^\perp$.
5. Define $f : \mathbb{C}^3 \to \mathbb{C}$ by $f(a,b,c) = (a+b+c)/3$. Find a vector $y \in \mathbb{C}^3$ such that $f(x) = \langle x, y\rangle$ for each $x \in \mathbb{C}^3$.
6. (*Riesz Representation*) Let $f : V \to \mathbb{F}$ be a linear functional on a finite dimensional ips V. Prove that there exists a unique $y \in V$ such that $f(x) = \langle x, y\rangle$ for all $x \in V$.
7. Let $A = [a_{ij}]$ be an $m \times n$ matrix with $a_{ij} \in \mathbb{C}$ and $n > m$. Show that there exists a nonzero vector $(\alpha_1, \ldots, \alpha_n) \in \mathbb{C}^n$ such that $\alpha_1 a_{i1} + \cdots + \alpha_n a_{in} = 0$ for all $i = 1, \ldots, m$. Interpret the result for linear systems.
8. Suppose a linear homogeneous system $Ax = 0$ has m equations and n unknowns. Show the following:
   (a) If $\operatorname{rank}(A) = n$, then $Ax = 0$ has a unique solution, the trivial solution.
   (b) If $\operatorname{rank}(A) < n$, then $Ax = 0$ has infinitely many solutions.
   (c) If $m \ge n$, then both (a)-(b) are possible.
   (d) If $m < n$ then $Ax = 0$ has infinitely many solutions.
   (e) $Ax = 0$ has infinitely many solutions iff the columns of A are linearly dependent.
9. Let the linear system $Ax = b$ have m equations and n unknowns. Show the following:
   (a) If $\operatorname{rank}([A\,|\,b]) > \operatorname{rank}(A)$, then $Ax = b$ has no solutions.
   (b) If $\operatorname{rank}([A\,|\,b]) = \operatorname{rank}(A) = n$, then $Ax = b$ has a unique solution.
   (c) If $\operatorname{rank}([A\,|\,b]) = \operatorname{rank}(A) < n$, then $Ax = b$ has infinitely many solutions.
   (d) $Ax = b$ has at least one solution iff $b \in \operatorname{span}\{A_1, \ldots, A_n\}$.
   (e) $Ax = b$ has at most one solution iff $A_1, \ldots, A_n$ are linearly independent.
10. Let $A \in \mathbb{F}^{m\times n}$. Let $U = \{X \in \mathbb{F}^{n\times k} : AX = 0\}$. What is $\dim(U)$?
11. Find least squares solutions for the following systems:
    (a) $3x + y = 1,\ x + 2y = 0,\ 2x - y = -2$.
    (b) $x + y + z = 0,\ -x + z = 1,\ x - y = -1,\ y - z = -2$.
12. Let $A \in \mathbb{F}^{m\times n}$ have linearly independent columns. Show that if A has a QR-factorization where the upper triangular matrix R has positive diagonal entries, then that is the only possible QR-factorization.

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